

As you can see, there isn't too much to do here. The themes listed in the box aren't as comprehensive as the themes in WindowMaker—they just change a few aspects of windows, such as borders, title bars, etc. You can download extra theme files and install them using the 'Install New Theme' button. A little more customisation can be achieved by changing

the title bar font. The last two options allow you to determine how the active window is chosen and how many virtual desktops are available. The default is four virtual desktops.



Metacity configuration offers limited customisability

Switching into Metacity within KDE

Even if you are a Gnome fan, you might find yourself using KDE, especially with the sheer number of applications that are bundled with it. KDE comes with its own window manager, kwin, which is tightly integrated into the desktop. You could still replace kwin with Metacity, although



Two Desktop Environments running simultaneously

it is slightly tricky.

To do this, open a terminal on your desktop and type the following command:

```
ps -ax | grep kwin
```

This will give you an output similar to the one given below

```
2701 ? S 0.00 kdeinit: kwin
```

The first number gives the process ID of the kwin window manager; you can use this ID to kill the window manager by typing the following command in the terminal:

```
kill -9 2701
```

The last number in the preceding line—2701 in our example—is the process ID of kwin. After killing the win-

Processes in Linux

What is a process?

Any program invoked on your Linux machine runs as a single process or a group of processes. Every process is assigned a unique process ID that is used by the kernel to identify it. To get a snapshot of all the current processes and their process IDs, use the command **ps -ax**

Killing a Process

It's possible that a process may not terminate as expected or you may want to manually terminate a process.

To kill a process within Linux, obtain the process ID and type in **kill pid** into a terminal.

This should kill the process. However, if a process ignores the kill command and does not get terminated, you would need to enforce the kill command. For this, type in **kill -9 pid**

The -9 flag here ensures that the process is terminated, as the it cannot ignore this kill signal from the kernel.

dow manager, you will notice that all the colourful borders of the opened windows have disappeared and you are unable to move them around. You can still use them—this is because the desktop is still alive and running, but it cannot figure out how to go about handling the windows since kwin, the window manager, has been killed and no other window manager is running. Now choose Run Command... from the start menu, and type **'metacity --replace'** in the dialog box that comes up. Click on Run and you will have Metacity managing your open windows under KDE.

Running KDE inside Gnome

Open the Run Command... dialog from the start menu and type **'startkde'**, which starts the K Desktop Environment inside the Gnome desktop environment. You'll get two different desktop panels, one above the other, from where you can access the start menus and applets of both KDE and Gnome. This can be of help when you need to quickly use the KDE apps straight from the KDE taskbar and start menu.

However, if you're running two desktop environments, you will need a lot of memory or else your Linux box is going to slow down to a crawl! ■

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About Managers and Environments

Here's some information to help you understand the differences between window managers and desktop environments.

The X server, commonly called X, is a portable GUI used on several platforms and systems, but mostly associated with UNIX. There are several implementations of X available, with the free version-XFree86-used for Linux. X gained wide acceptance with its eleventh revision, famously known as X11.

X achieves its complete functionality through the working of several independent components, rather than a single monolithic—and thus hard to manage—structure. The most significant advantage of such a modular structure is its customisability.

The individual components of X can be tweaked and changed by users and this is where the window managers come in.

Since users can choose to change the look and feel as well as the interaction of windows, a streak of individualism can be added into what is essentially a very com-

monplace product. You could choose to implement a new style just as often as you want to!

This is where the window manager comes in. The window manager manages the windows, and provides the interface where the users can interact with various application windows and defines the look and feel of your workspace.

Desktop environments provide a complete user interface and come bundled with several applications and utilities, eg. GNOME and KDE (K Desktop Environment).

Window managers can work without desktop environments, but the reverse is not possible. This happens because desktop environments always need a window manager to manage their windows, while window managers can draw and manage their windows by themselves. Sounds like a reason to junk that heavy desktop environment? Hold on, you forgot all about those wonderful bundled applications!